

Whither engineering ? — thinking for the future

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Must the future always arrive by accident? Technologists are good at studying hardware and software when designing new products and services. Perhaps equal or greater insight could be found if technologists examined themselves and their assumptions, by thinking about thinking? Mental flexibility or rigidity can affect both the detail and scope of decision making, especially in uncertain or rapidly-evolving situations. If you want to invent the future, do you need luck, intuition, experience or genius? Is such talent innate or can anyone learn the skills involved?

1. Introduction

One of the world's oldest books is the Chinese 'Book of Change', which describes ways of foretelling the future. In the light of a new millennium, with technical change all around, futurology is again a popular if not entirely respectable subject. You have to be both talented and brave to put 'Futurologist' on your business card. Astrology and spiritualism might seem too close for comfort. Some people would be tempted to read 'futurologist' and think of 'charlatan'. Why should futurology ever be taken seriously? Surely no one can predict the future; but futurology does not mean any kind of clairvoyance or precognition.

In this paper, futurology means two things. Firstly, it means a kind of mental preparedness, to be less frequently caught out when the unexpected happens. More controversially, it also means a kind of speculation that exploits both objective and subjective, or even subconscious, ways of thinking. New bio-medical technology has begun to offer hard evidence for the objective validity of soft kinds of thinking.

This second kind of futurology is like advertising, because it is neither dependable nor useless. It has always been possible to do more than react to change as it happens — markets can be led. The motive is money, because fortune favours the prepared mind. Many factors suggest revenue growth in telecommunications, including the popularity of the Internet and the growing importance of timely information to all walks of life. Other developments imply a decline in revenue, such as the tariff reductions spurred by increased competition and deregulation.

Market analysts have tried to predict the outcome of these complex and contrary factors. A layman might say that telecommunications would grow at the same rate as the economy, if the role of the industry remains the same. The commercial application of futurology is to search for new factors that might increase the role of communications, as discussed by Walker [1] and Greenop and Williams [2].

Several famous names in futurology make their motives explicit, as in recent books by Schwartz [3], Negroponte [4], Rawlings [5], Plant [6], Barrett [7] and Dertouzos [8]. These books are serious, yet too populist for full-blown scientific respectability. It has been said that the best way to foresee the future is to invent it. If so, perhaps it is time for some academic respectability in futurology.

There is little really new in what follows. In all sorts of places, people already do what this paper describes, though perhaps in accidental or unconscious ways. The paper starts with a light-hearted look at some popular and less popular approaches to futurology. There follows a more serious look at modes of thought and how they relate to the interactions of technology and society. Some potential implications for research and development are then discussed and finally some suggestions are made about the skills needed for technical futurology.

The paper is a plea for a more deliberate understanding of how we go about thinking, and for a wider recognition of the value of freedom of thought. If revolutionary change in the social impact of technology does lie ahead, as many commentators suggest, the skills that underlie good

futurology will be relevant to a much wider audience than just researchers and technologists.

2. Style and method

2.1 Outlook

In the mass media [9], three styles of popular futurology are common, labelled in Fig 1 as the views of economists, technologists and the prophets of doom. The economist style tends to forecast a gradually evolving continuation of what has already happened. These predictions are short-term and conservative, seeing change as a matter of degree not direction.

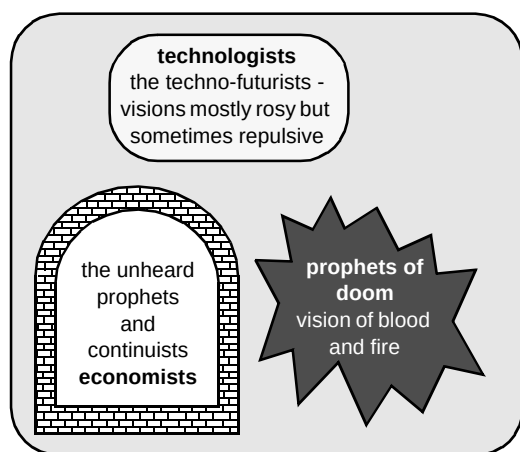


Fig 1 Three kinds of futurologist outlook.

The technologist style looks further ahead, to an irrepressibly sunny future that has lots of new toys. These predictions can seem plausible, but they can also fail comically with a silliness that is entertaining. One such vision, from the 1950s, saw us surrounded by domestic robots and flying to work in helicopters by the turn of the century. The technologist style of popular forecast often sees a future where only the technology really changes. Social attitudes or cultural values change little, if at all, in these predictions.

More far-reaching and significant change has come from the social impact of technology that was already available in the 1950s. For example, telecommunications has changed the nature of work and broadcasting has fostered greater social equality. Only the prophets of doom are bold or foolhardy enough to suggest radical change in life-style or social organisation. Their forecasts commonly see a triumph of the very worst in human nature, whatever the technology. This is why their arguments are often seen as opposing technical change.

All three styles of popular futurology make forecasts that are clear, uncomplicated and unambiguous, dealing with a rational world of facts from a logical point of view.

Yet none of these mass-media styles of forecasting has a good track record of foresight.

2.2 Difficulties

The great difficulty, in predicting any one aspect of the future, is that everything else can change too [10]. Interactions can occur between areas that seem unrelated. If these interactions are not linear, the effects of chaos quickly replace the workings of logical determinism. Accurate forecasts of the future are then impossible.

A second difficulty, in finding even partial success in technical foresight, stems from interaction between technology and society. Minor changes of cultural attitude or social value seem to be at least as influential, over the long-term, as major changes in technology. Yet quite minor aspects of technology can stimulate or catalyse unexpected change in social or cultural values.

Cultural values are hard to measure and theories of society are controversial. Social, cultural and human issues remain resistant to explicit logic. Futurologists in this field need subjective skills, like judgement, experience, insight and wisdom, and they suffer from subjective deficiencies like prejudice and bias. The outlook for academic respectability in technical futurology might look bleak.

2.3 Other approaches

But there are other, and more serious, approaches to futurology. University courses at Leeds and Oxford offer systematic study of foresight, while Houston/Clear Lake and Hawaii Universities in the USA have extensive involvement with studies of the future [11–14]. Organisations like the World Futures Studies Federation and the Institute for Alternative Futures have provided planning and policy-making advice about health, education, justice, communications and the environment to governments and major companies. In all these serious approaches, flexibility of mind plays a fundamental role. Rehtin [15], in speaking about the creation of complex, first-of-a-kind systems, recommended that those involved should ‘step back and think about the best ways to think’.

Schwartz was working in the oil industry when he came to study futurology, as a tool for coping with the threats and opportunities of radical or uncertain change. His book [3] describes his scenario-planning technique, based upon intuitive and intangible information rather than objective measurement and analysis. He argues for contingency planning that includes some semi-crazy as well as semi-sensible scenarios. This is the ‘mental preparedness’ kind of futurology.

The management theorist Handy [16] also seems to adopt an intangible approach. He examines the nature and

origin of the frame of mind that might begin to ask the right sort of questions about the future. His approach is more concerned with how to think about the impact of new events, and their motivations, than with the detailed internal workings of those events.

One of the most successful pieces of futurology this century came from a science fiction writer. Clarke's proposals about geostationary communication satellites [17] now look like invention not foresight. But in 1945 these ideas were pure dreams, because implementation was impossible. Today, Clarke's proposals about building elevators that rise all the way into space [18] seem just as implausible. Does that mean they cannot be built?

The best science fiction often makes logical and self-consistent explorations around imaginative (and sometimes impossible) leaps in culture, science or technology. Such an approach has the advantage of separating metaphor from manifestation and encouraging imagination. Gene Roddenberry's *Star Trek* used this approach to gain a reputation for social prediction, challenging taboos of race and culture.

3. Modes of thought

3.1 Two worlds

Serious futurology relies on impressions and opinions from the world of subjective thought (Fig 2), as well as on techniques from the objective world of science and technology. A formal separation of these two modes of thought has a long history in the West, going back to the Greeks and beyond. Galileo emphasised it, in trying to avoid accusations of heresy for saying the Earth went round the Sun. In the 18th century, the politician Burke promoted it, apparently because he feared that scientific method might steer political debate [19]. Earlier this century, the formal separation widened into an educational schism between the arts and the sciences — what C P Snow called 'the two cultures'.

This schism is deeply entangled with cultural attitudes about science. In the 19th century, it was not disreputable for a scientist as eminent as Sir Oliver Lodge to also be a keen spiritualist. In this century, the public's initial reaction

to radio was a mixture of disbelief and fear. To conjure disembodied voices out of thin air with special crystals seemed closer to seance than science. Because science broke its links with alchemy and magic relatively recently, subjective thinking in science can still have disreputable associations today.

3.2 Re-unification

Yet in recent years, it has become increasingly obvious that separating mental processing into distinct objective and subjective domains is not only needless but unproductive [19, 20]. It seems ironic, despite the prodigious success of objective methods in testing scientific ideas, that creating those ideas often takes subjective, non-rational or 'hidden' kinds of thinking. If artists are free to use logic as well as intuition, then surely technologists can use intuition as well as logic. In ordinary life, both modes of thought are used seamlessly and without guilt.

The psychologist Claxton [21] laments a requirement that is common in education — 'show your working' — because it heaps shame on intuitive thinking, whether insightful or not. He points out that new techniques for measuring brain physiology suggest a deterministic basis for some kinds of subjective thinking.

Experiments with subtle bias, added to otherwise random action in computer games, have shown that we can act on information we do not consciously know we have acquired. It is possible that some kinds of intuitive decision are not purely personal, but come from detecting patterns in chaos — that is, in behaviour that superficially seems random but actually has a deterministic origin.

Goleman [22] says the conventional view of intelligence is too narrow, devaluing the crucial role of emotion and the subconscious in human development. The journalist Holderness [23], talking about the changing nature of knowledge in a networked world, says: 'The skills we now need to survive, to ride the flood-tide of data and not drown, are curiosity, analysis and sceptical evaluation. Education is no longer — and never was — about facts.' Such outlooks bridge the objective and subjective worlds, returning to an earlier tradition of education in the round.

3.3 Using the subjective

Handy's 'frame-of-mind' approach is less concerned with the presence of positive facts, in finding support for a case, than with establishing the absence of negative evidence. It is more connected with the present day and less immersed in any specific future vision. The approach leads to broad conclusions that tend to contain ambiguities, so that specific detail is difficult to extract. General directions and areas of possibility are what emerge.

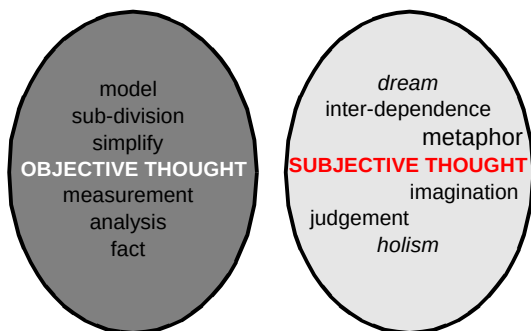


Fig 2 Two worlds of thought

At first sight, this approach might seem too foggy-minded or unambitious to be of much use. But for futurology, this kind of approach can be advantageous because it is motivated by a pragmatic question — what is the best way to get to the future from here? By being less ambitious, the conclusions are less likely to be wrong-footed by events. This kind of thinking avoids technical detail and is closer to the kind of abstract or multi-disciplinary thinking that has been the hallmark of good futurology in the past.

Some examples of the issues suggested by a more intangible or exploratory approach to thinking are shown in Table 1, contrasted with the more traditional, specific kind of approach.

Table 1 Examples of specific and exploratory thinking in technology.

New technology	Specific	Exploratory
Larger visual displays	More pixels for sharper pictures Greater viewing distance with same clarity Higher costs Reduced portability	Altered purpose, new uses or use in new places Communicating more with feelings than facts Different social reactions Change of metaphor
Computers that talk and listen just like a person	Greater access to computer data Increased traffic volume for new services Greater convenience and ease of use Lower operating costs	Change in human style of speaking in telephony Less trust in telephone conversations More trust in validity of computer data Legislative or regulatory controls Change in working habits

But the strongest justification for the greater use, in concert, of both objective and subjective modes of thought comes from the changes currently under way in the interaction of technology and society. Nowhere are these changes more obvious than in the convergence, and collision, of telecommunications, computing and entertainment.

4. Technology and society

4.1 Four phases

A simple picture of the interactions between technology and society divides developments into four phases (see Fig 3). The classification depends on the amount of interaction and on whether social or technological factors provide the impetus or motive source for change. The phases with strong interactions are where futurology is useful.

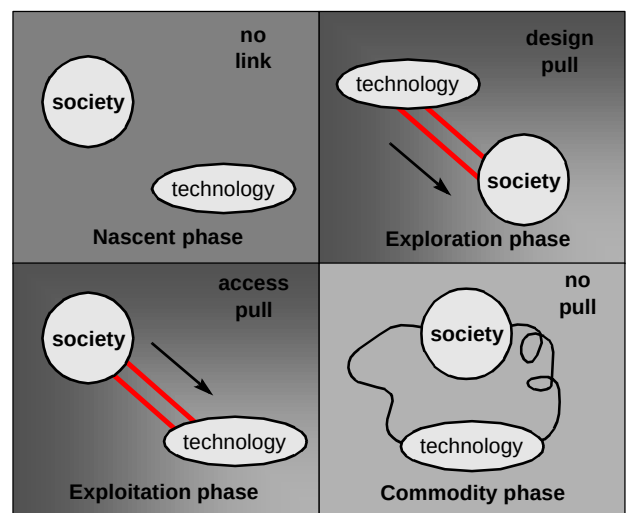


Fig 3 Four phases of socio-technological interaction.

In the nascent phase, there are no interactions and technological development is steered by other factors. An example of this phase is the early history of clocks. Most people took their time from the sun and the steadily increasing accuracy of clock design had no social impact for centuries. It was the developing needs of marine and land transport that gave a social impetus to the development of clock technology, and moved that technology out of nascent phase conditions.

Stability is also the characteristic of the commodity phase, at the bottom right of Fig 3, where interaction between technology and society is not strong and occurs slowly. One example is the present state of 'pen-and-paper' technology. Hand-made drawings have a wide range of varied yet quite stable social purposes, that are little influenced by new technology in dye or paper production.

The exploitation phase has a partial kind of stability, though its interactions can be energetic. This phase is dominated by the refinement and expansion of existing ideas and applications, where technology provides the motive source for change. Here, the social purpose of a technical idea has been established and technological development creates changes that are mainly quantitative — more, better, faster. Futurology is primarily technical in this phase and is relatively precise. Moore's law is probably the most famous example of this kind of exploitation phase futurology.

The exploration phase is characterised by instability and uncertainty. Social needs provide the motive source for change and define the direction of development of new technical ideas. New things become possible in exploration phase conditions, so change is mainly qualitative and futurology is not primarily technical. In this phase, journalists and artists can have more interesting things to say than technologists can, although their predictions may be diverse or extreme. Examples of this phase are the

invention of mechanised printing, where society's appetite for political gossip funded the development of better printing machines, or the invention of radio, where commercial and military applications funded a flowering of electronic technology from the mid 1920s to the late 1940s.

This classification of the exploration and exploitation phases might seem counter-intuitive or the wrong way round. Technology is conventionally thought to pull or disturb society when a new invention occurs. Equally, society is often seen as controlling the exploitation of mature technology. This conventional outlook confuses action with intention. The driving or motive source, that supplies the decisions and controls the enabling factors in the use of technology, is what dictates the course of socio-technical interaction.

Some types of communications technology, such as mobile telephony, can be linked with more than one phase. Drawing a clear distinction of phase in some developments can be difficult, even with the benefit of hindsight. Despite this, the classification does lead to a useful insight about the different kinds of thinking needed to develop technology.

4.2 *Thinking about thinking*

The social purpose and cultural impact of technology is established in exploitation phase conditions. Everyone knows what the technology is for and the problem is not what to do but how to do it. Here, objective and analytical thinking dominates and scientific development is energetic. There is little room or need for techniques from the subjective world of thought.

In exploration phase conditions, the opposite is true. Social purpose and cultural impact are volatile and unexplored. No one fully knows how the technology should be used and the problem is not how to do it, but what to do. Subjective thinking plays a dominant role and technical development does not follow an obvious or smooth path. The early history of the telephone is a good example of exploration phase conditions, where enthusiasm and imagination drove practical deployment — despite a profound lack of objective and scientific knowledge about the technology.

4.3 *A change of phase?*

Engineering has been defined as the art and science of making something useful. In exploitation phase development, the boundaries that are pushed back are the boundaries of science and technology. The science has long been understood, the art well practised and the usefulness established beyond doubt in telephony and television.

For many decades, global research and development in telecommunications has been dominated by exploitation,

providing more of the same at lower cost with better performance. The same is true of television. A time traveller from 1937, who had seen early BBC broadcasts, would not be greatly surprised by television today. The idea remains the same, though its social impact has expanded. Better pictures, in colour, with more channels at lower prices are the result of sustained exploitation. Both telephony and television show the characteristics of socio-technical interaction in the exploitation phase, where change is a matter of degree and not direction.

This is not true of computers. A time traveller from 1967, who was well acquainted with computers, their architecture and applications at the time, would be mystified by the multi-faceted nature of desktop computing in the home today. Imagine the reaction — 'Why is this called a computer? It isn't computing anything. Where have all the numbers gone? How could computing have led to this?' In the last decade, the desktop computer industry has seen rapid change in its social impact and much more seems yet to come. This is typical of the conditions of exploration, where all the engineering boundaries get pushed back at some time — in art, science and usefulness.

Many commentators have suggested that telecommunications and television will soon experience a similar kind of change in their social impact. A change in the nature of change would then occur, with a shift in phase of socio-technical interaction.

5. **The nature of change**

5.1 *Different shifts*

The Internet has been cited as causing just as dramatic a shift to exploration conditions, since it is a new idea with unknown social impact. Commentators like Toffler [24] and Plant [6] have suggested a more extreme shift, with radical changes to the social norms of education, work and leisure. They see the familiar institutions, values and practices of these areas merging and mutating. Toffler sees 'knowledge-worker' transactions replacing the dealings of a 'smoke-stack' economy. Davis and Meyer [25] foresee a world that blurs the conventional differences between products and services, buying and selling, employer and employee, customer and company. In these kinds of future, telecommunications plays an even larger and more central role than it does today.

A few commentators have suggested a different shift, to commodity phase conditions, where the role of technical innovation becomes minor after market saturation is reached for both television and telecommunications. Here, prices fall dramatically and choices stabilise, without substantial change in the social role of either industry. In this kind of future, the role of communications stays the same in an unchanged cultural and social world.

All commentators seem to agree that what is changing most of all in telecommunications is the nature of change. Change used to mean more, better and faster kinds of familiar things. Most futurologists now see change as increasingly turbulent and dominated by difficult questions for both telecommunications companies and society — questions about when, where, how much, why and what.

5.2 Research and development

A shift towards the exploration phase of socio-technical interaction would have significant implications in commercial research and development for telecommunications. Figure 4 shows the path, from conception to deployment, that is typical of exploitation phase development. The path is essentially linear and new technology can be successfully devised and designed in isolation.

In contrast, Fig 5 shows a very different kind of path, typical of exploratory phase development, where user reaction is tangled in loops with technology and financial limits. Pilot tests, user trials and other kinds of social feedback grow in importance and become relevant earlier in the design cycle, when social purpose steers exploration. Subjective and human factors become closely entangled with objective design decisions, rather than a separate discipline, in exploration phase conditions.

Walker et al [26] provide a good example of exploratory phase development, where playful experiment (sometimes thought of as ‘smoke and mirrors’ from an exploitation phase viewpoint) has a vital role in the joint progress of technology and new social purpose.

Public testing of new ideas has sometimes been seen as disadvantageous in telecommunications, because of the risk

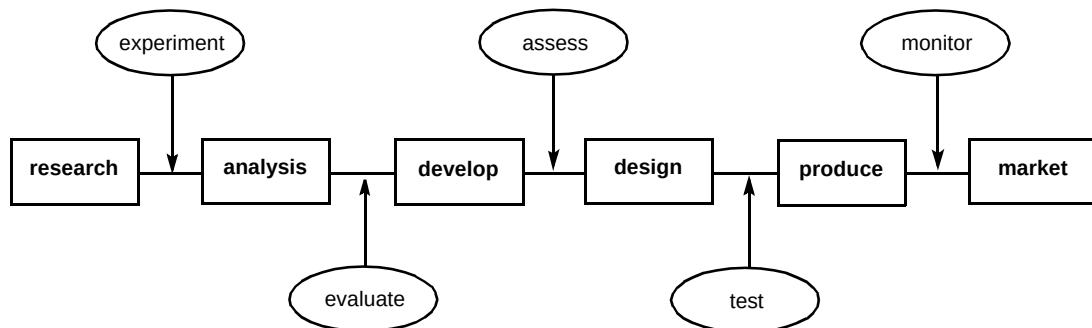


Fig 4 Path of technology development in exploitation phase conditions.

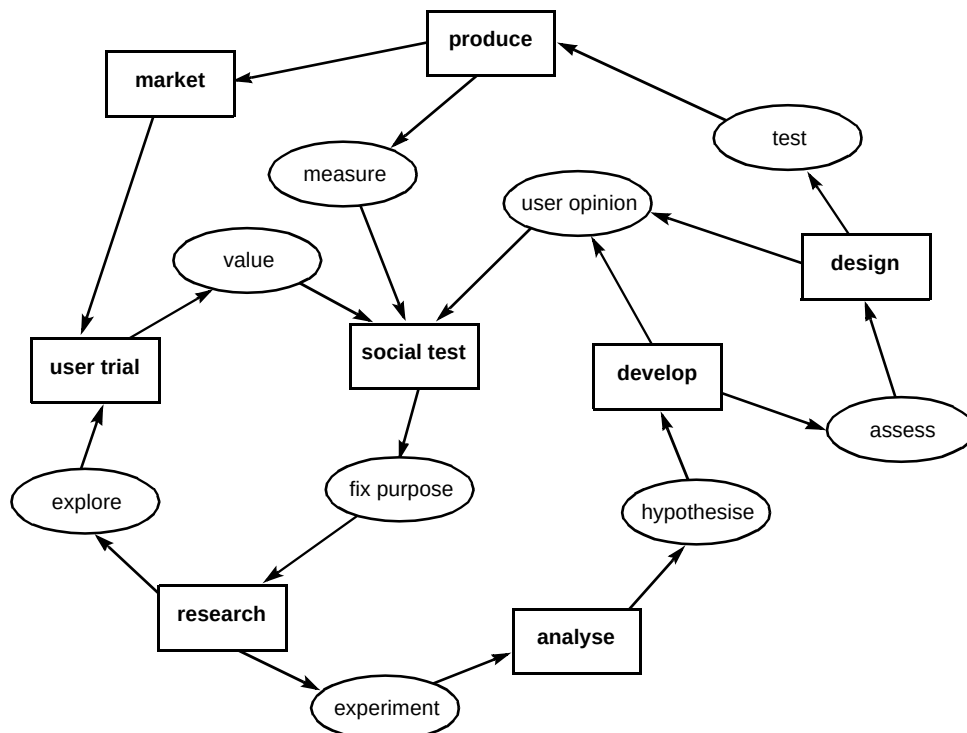


Fig 5 Paths of technology development in exploration phase conditions.

of negative publicity from unsuccessful trials — the typical result of an exploitation phase outlook.

5.3 Social change

Of course, telecommunications is just one part of a much wider tide of change. Sir Peter Ustinov, in his preface to Lazlo's book '3rd Millennium' [27] says:

'We live in the midst of the fastest and most profound transformation in the history of humanity... Living and acting in the new condition calls for radically different ways of thinking and acting... To live in the third millennium we shall need more than incremental improvements on our current rationality; we shall need new thinking joined with new ways of perceiving and visioning ourselves, others, nature and the world around us. We shall need new values, ethics, and modes of life and behaviour.'

Dertouzos, head of Computer Science at MIT, presents a comprehensive study [8] of the potential social and economic impacts of information technology. He gives five agendas for action, customised to different interest groups. He is an enthusiast for technology, yet also sceptical and undazzled by its potential. He includes an exploration of what might happen to human relationships and the human psyche — leading to what he calls 'a renaissance in the way we view ourselves'. His book is a good example of the power of unifying objective with subjective thinking, of the strength that can come from joining the technical with the humane.

6. Cultivating attitudes

6.1 A difference of outlook

Two different kinds of futurology, with different skills and intellectual attitudes, seem to be appropriate to different phases of interaction between technology and society. Exploitation phase futurology needs technical and scientific analysis and extrapolation. In contrast, exploration phase futurology needs artistic and abstract qualities, like fantasy and philosophy, that are seldom associated with technical or scientific knowledge. The risks of futurology are much greater in the exploration than in the exploitation phase, but so are the rewards.

The technical and popular media seldom draw a clear distinction between these two very different kinds of futurology. This lack of differentiation can seriously hinder understanding of the impact of technology. It can also create confusion, especially when changes in the exploration, exploitation and commodity phases are all happening simultaneously. The 'chalk-and-cheese' difference between exploration and exploitation conditions is fundamental. It is possible to eat chalk and to write on a blackboard with

cheese, but it is not satisfying and makes a terrible mess. Informed debate about future possibilities depends on fully understanding the basis of the speculation.

For example, data traffic from new technology and emerging applications is currently growing so fast that it may soon exceed the volume of traditional voice traffic. What are the characteristics of this future data? Data traffic is commonly described as bursty, time-insensitive and asymmetric in bandwidth.

Yet this is the wrong answer, because it is the wrong question. The way the question is framed betrays an outlook rooted in the conditions of the exploitation phase, where the social purpose of technology is well known. Yet many of the new developments that are driving data growth are currently located in the exploration phase, where their social purpose is poorly understood. From an exploratory phase viewpoint, asking the characteristics of data is just as sensible as asking whether water is inherently round or square. The characteristics of future data traffic depend strongly on the social purpose and popularity of those new technologies and applications that will create its growth.

6.2 Exploring attitude

The author suggests that, for exploration phase conditions, futurology should adopt the 'pragmatic centralist' kind of viewpoint (Fig 6). This borrows from the three outlooks in popular futurology described in section 2.1, but also employs subjective thinking. The intent is to use some of the caution of economists to display less drama than has been traditional in technical prediction, with more of the analysis and exploration of social interaction favoured by the prophets of doom. Stentiford [28] says that endorsing this position is equivalent to recognising the value of greater freedom of thought.

This viewpoint reunifies the two worlds of thought, yet it lacks the comfort and status of either world. The 'pragmatic centralist' outlook seeks channels between established areas of objective and subjective thinking. A

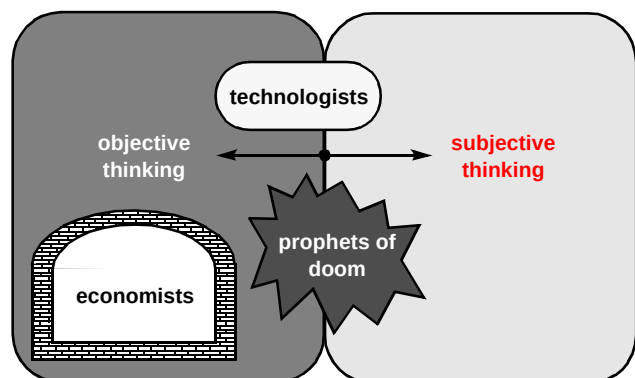


Fig 6 'Pragmatic centralist' viewpoint.

lack of knowledge about, and use of, such channels may be the greatest disadvantage created by the conventional schism in modes of thought.

6.3 *A personal list*

If exploration phase futurology needs a different frame of mind, what sort of mental attitudes or qualities are needed? The author's personal and partial list is shown below. Readers are invited to make their own list. No one person is likely to be strong in all these qualities and listing them is like trying to define sainthood — it is notable that these attitudes are not necessarily those conventionally associated with engineering or business success:

- lack of a fixed world view, open mindedness, neutrality,
- doubt, healthy scepticism,
- humility,
- awareness of pros and cons of multi-disciplinarity,
- searching for, and challenging, preconceptions and assumptions,
- knowledge of humanities and science,
- historical perspective,
- people-centred analysis and synthesis,
- imagination, creativity, fantasy,
- enthusiasm, vitality, freshness,
- fascination, curiosity.

Some of these attitudes demand the experience of an adult, but it is no accident that other attitudes seem child-like, because both ignorance and innocence give a freshness of view. The most important attitude in the list may be humility, if only because it seems to underpin the others. Handy [16] claims that humility is the most important thing that modern science has given the world, by proving just how much we do not know.

7. Conclusions

Trying to predict the future is a disreputable activity. It does not seem compatible with the methods or culture of science and technology. Yet commercial research and development is necessarily involved with future possibilities. The author suggests that futurology is akin to advertising, because it is neither deterministic nor a fraud. Futurology can be worth doing even though success is not guaranteed.

Two different kinds of futurology have been identified, with significantly different characteristics. The first, related

to market analysis and economic forecasting, is suitable when developing technology that has an established social purpose. The second, related to imaginative speculation, is needed when developing technology with a poorly understood social impact. These two types of futurology require very different mental attitudes and intellectual skills.

Yet in common usage, these two kinds of futurology are mixed, merged or not clearly differentiated. This damages the credibility of futurology and greatly reduces the understanding of its role. Journalists may indulge in future fantasies for entertainment or diversion, but technical futurology has a serious, if under-appreciated, purpose. It can help shape, in different ways, the priority and structure of commercial research and development.

Many commentators, from Toffler to Handy, have predicted change in the nature of change, arising from the social impact of new technology. If these predictions are even partly true, the consequences are going to be dramatic. This paper has suggested why a change in the nature of change would lead to an increased role for the second kind of futurology.

This also would lead to a shift in the intellectual abilities and aptitudes needed in the planning, execution and assessment of some types of research and development in electronic technology. Objective measurements and rational analyses would retain their importance, but subjective abilities like imagination and mental playfulness, that operate outside the bounds of strict logic, would play a greater role than has been traditional. Fresh thinking would be needed, not just in the business world as Handy and Toffler argue, but in the world of engineering too.

It is conventional to separate objective and subjective thinking, but this becomes counter-productive when these issues become inter-dependent. Subjective modes of thought are commonplace in everyday life — yet they are seldom used openly in the analytical world of science and engineering. The contemplative mind seems fragmented and irrational, because its workings are inaccessible. But there is growing objective evidence, in experimental cognitive science, for the mechanistic validity of intuition — for those thoughts and decisions that seem to come from nowhere. The brain seems to be able to extract useful information from some kinds of chaotic and confusing situations, without consciously knowing how.

Many instances of dazzling genius and pure luck can be found in the history of technology, especially at times of unexpected change. But that does not seem like a good reason to wait for either luck or genius. It does seem to be worth thinking about the way in which we go about thinking. The attitudes, skills and understanding that support the speculations of good futurology look to be both undervalued and learnable.

Furthermore, the social impact of new technology is likely to require these skills from a much wider population, to cope with turbulent change. It may sound strange, but futurology has lessons for us all — only more as technique than prophesy.

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